

Jacob Chalif

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EDUCATION

- **Dartmouth College** 2024 – 2026 (*expected*)
M.S. in Earth Sciences Hanover, NH
 - Advisor: Erich Osterberg
 - Thesis: 1,200 years of Siberian wildfire preserved in an Alaskan ice core
- **Dartmouth College** 2017 – 2021
B.A. in Environmental Earth Sciences Hanover, NH
 - GPA: 3.84/4.00 (Magna Cum Laude)
 - Senior thesis: Effect of Mid-20th Century Jet Stream Waviness on the U.S. “Warming Hole”
 - Advisors: Trevor Partridge and Erich Osterberg

RESEARCH EXPERIENCE

- **Dartmouth Ice, Climate, and Environment (ICE) Lab** 2021 - Present
Lab Manager; Research Assistant Hanover, NH
 - Extensive experience in continuous flow analysis (CFA) ice core melter systems and ice core impurity geochemistry (dust, sea salts, sulfur species (MSA, SO₄), heavy metals, fire tracers, water isotopes)
 - Led projects measuring fire aerosols in the Denali ice core, re-interpreting Arctic MSA variability using atmospheric chemistry box modeling, and improving our understanding of impurities in the world’s oldest (>1 million years) ice cores from the Allan Hills, Antarctica

PUBLICATIONS

- 4 Chalif, J.I., Osterberg, E.C., and Partridge, T.L. (2025). **A wavier polar jet stream contributed to the mid-20th century winter warming hole in the United States.** In *AGU Advances*. 6, e2024AV001399. DOI: 10.1029/2024AV001399. (Selected for an [Editor’s Highlight](#).)
- 3 Jongebloed, U.A., Chalif, J.I., Tashmim, L., Porter, W.C., Bates, K.H., Chen, Q., Osterberg, E.C., Koffman, B.G., Cole-Dai, J., Winski, D.A., Ferris, D.G., Kreutz, K.J., Wake, C.P., Alexander, B. (2025). **Dimethyl sulfide chemistry over the industrial era: comparison of key oxidation mechanisms and long-term observations.** In *Atmospheric Chemistry and Physics*. 25, 4083–4106. DOI: 10.5194/acp-25-4083-2025.
- 2 Henley, C.A., Meyer, C.R., Chalif, J.I., Hollmann, J.L., and Raskar, R. (2025). **Measurement of snowpack density, grain size, and black carbon concentration using time-domain diffuse optics.** In *Journal of Glaciology*. 71, e6. DOI: 10.1017/jog.2024.81.
- 1 Chalif, J.I., Jongebloed, U.A., Osterberg, E.C., Koffman, B.G., Alexander, B., Winski, D.A., Polashenski, D.J., Stamieszkin, K., Ferris, D.G., Kreutz, K.J., Wake, C.P., and Cole-Dai, J. (2024). **Pollution drives multidecadal decline in subarctic methanesulfonic acid.** In *Nature Geoscience*. 17(10), 1016–1021. DOI: 10.1038/s41561-024-01543-w.

PUBLICATIONS IN REVIEW/PREP

R=IN REVISION, S=SUBMITTED (IN REVIEW), PXX=IN PREP FOR SUBMISSION IN 20XX

- P26 Chalif, J.I., *et al.* **A 1200-year ice core record of southeast Siberian fire.** *Anticipated 2025 submission.*
- P26 Chalif, J.I., *et al.* **High-resolution impurity measurements from Allan Hills ice cores.** *Anticipated 2026 submission. (Likely multiple papers.)*
- P26 Koffman, B.G., Chalif, J.I., *et al.* **Local versus long-distance dust provenance in a 2019 shallow core at Mount Hunter, Alaska.** *Anticipated 2025 submission.*
- P26 Kirkpatrick, L., Chalif, J.I., *et al.* **A novel integration of ice and gas analysis to resolve fine layering in a basal-impacted million-year-old ice core.** *Anticipated 2026 submission.*
- P26 Hudak, A., *et al.* (including Chalif, J.I.). **Dust and methane measurements from Allan Hills ice cores.** *Anticipated 2026 submission.*

- P26** Choi, A., *et al.* (including **Chalif, J.I.**). **Dust source provenance in an Allan Hills ice core.** *Anticipated 2026 submission.*
- P26** Tipton, L., *et al.* (including **Chalif, J.I.**). **Causes of the Late Holocene Dust Spike in Antarctic Ice Cores.** *Anticipated 2026 submission.*
- P26** Brooks, H., *et al.* (including **Chalif, J.I.**). **A 1200-year ice core record of lead isotopes from the Denali, Alaska, ice core.** *Anticipated 2026 submission.*
- P26** Kirkpatrick, L., *et al.* (including **Chalif, J.I.**). **Denali ice core water isotopes record precipitation-weighted temperature of North Pacific.** *Anticipated 2026 submission.*

RESEARCH FUNDING

- 2026** Fieldwork support from the Dartmouth Changing Polar Regions cluster, Graduate Student Council, and Earth Sciences Department to drill ice cores at the Eclipse Icefield. Total grants: \$8,300.
- 2025** Sub-grant from COLDEX. High-resolution CFA chemistry analysis of Allan Hills ice cores. Erich Osterberg (PI), **Jacob Chalif** (co-PI). Total grant: \$35,000.
- 2024** COLDEX ECR Grant. Detailed layer analysis of oldest ice, phase 2. **Jacob Chalif** (PI). Total grant: \$5,000.
- 2024** COLDEX ECR Grant. Sulfur history of MIS6/5e deglaciation. **Jacob Chalif** (PI), Ursula Jongebloed (co-PI). Total grant: \$7,000.
- 2023** COLDEX ECR Grant. Detailed layer analysis of oldest ice. Liam Kirkpatrick (PI), **Jacob Chalif** (co-PI). Total grant: \$5,000.

FIELD RESEARCH

- **Eclipse Icefield, St. Elias Mountains (Yukon), Canada** *May – June 2026 (planned)*
 - Proposed research and secured funding to join a geophysics team from the University of Maine at the Eclipse Icefield
 - Leading shallow firn coring and surface geochemistry sampling and multiproxy laboratory analysis
- **Allan Hills, Antarctica** *Nov 2024 – Jan 2025*
 - Joined the Center for Oldest Ice Exploration (COLDEX) to drill ice cores in the search for the oldest ice. At 6.7 million years old, the ice we recovered is the oldest ice core ever drilled.
 - During 8 weeks at the Allan Hills, retrieved around 10,000 lbs of ice cores at various sites using 3 drill systems.
 - Also completed ApRES, ground penetrating radar, and GPS surveys.

PROFESSIONAL SERVICE

- **Peer reviewer:** AGU Advances, Environmental Research Letters, Climate of the Past, Journal of Geophysical Research: Atmospheres
- **Ice Core Early Career Researcher Worskhop (ICECReW) organizer**, 2024 and 2025 (lead organizer in 2025)

HONORS AND AWARDS

- **NSF GRFP Honorable Mention**, 2025
- **Magna Cum Laude**, Dartmouth 2021
- **High honors in Earth Sciences**, Dartmouth 2021
- **Warren F. Upham 1871 Geology Award for Outstanding Senior Thesis**, Dartmouth 2021
- **3 citations for academic excellence** during Dartmouth undergraduate coursework: (1) Introduction to Programming and Computation, (2) Climate Dynamics, and (3) Geomorphology

SCIENCE OUTREACH

- **Climate of Hope / Project 2100:** In February 2026, invited to give one of two keynote addresses at the Chicago Project 2100 conference for high school students and teachers (the other keynote speaker was Richard Alley). Additionally, I organized and ran activities at the related Climate of Hope event specifically for teachers.
- **Middle schools:** In April 2025, spoke with middle schools in New York (remotely) and Vermont (in-person) about Antarctic field work and ice cores, including demonstrations with real Antarctic ice samples
- **School of Ice:** In June 2023 and June 2025, designed/taught a lesson on ice core CFA techniques and chemistry analysis (which featured melting an ice core from Greenland and interpreting results in real time) for educators at minority serving institutions
- **Great Issues Scholars:** In 2023, 2024, and 2025, led a session for Dartmouth's Great Issues Scholars program on how ice core science informs 21st century climate change
- **Media outreach:** [Grist](#), [Boston Globe](#), [Eos](#), [New Scientist](#), [Süddeutsche Zeitung](#) (German), [Newsday](#), [NHPR](#) (also [Valley News](#) and [Concord Monitor](#)), [VT Digger](#), [WCAX](#) ([one](#) and [two](#)), [WHDH](#), [Newsweek](#), [Vermont Standard](#), [Earth.com](#), and [Dartmouth News](#) ([one](#) and [two](#))

SKILLS AND CERTIFICATIONS

- **Programming Languages:** Python, Matlab, Julia, R, C, ArcGIS/QGIS, HTML web design
- **Research Skills:** Large-volume climate data management (reanalyses, global climate/chemistry model output), geochemical analysis (SP2, particle counters), ice core melter systems, microprocessors, DC circuit design and solar power
- **Medical training:** WFR (exp. 07/2028), CPR (exp. 07/2027)

LOCAL CLIMATE AND ENVIRONMENT ENGAGEMENT

- **Sustainable Woodstock:** volunteer and trustee (treasurer), 2023-present
- **Upper Valley Adaptation Workgroup:** member, 2023-2025

UNDERGRADUATE STUDENT MENTORING

- Kat Adelman, Dartmouth'21, lab assistant (2 terms)
- Victoria Hofner, Dartmouth'22, lab assistant (2 terms)
- Liam Kirkpatrick, Dartmouth'22, research & lab assistant (3 terms)
- Abigail Lambert, Dartmouth'22, lab assistant (2 terms)
- Eric Youth, Dartmouth'22, lab assistant (2 terms), also advised senior honors thesis
- Parker Jones, Dartmouth'24, lab assistant (1 term)
- Chloe Nicolao, Dartmouth'24, lab assistant (1 term)
- Cady Rancourt, Dartmouth'24, Sophomore Science Scholar, studying "weather whiplash"
- John DeForest, Dartmouth'25, lab assistant (1 term)
- Maggie Druckenmiller, Dartmouth'25, lab assistant (1 term) and UGAR funded project (1 term)
- Erica Simon, Dartmouth'25, lab assistant (1 term)
- Laura Wilson, Dartmouth'25, Lab assistant (2 terms)
- Albert Zhang, Dartmouth'25, lab assistant (1 term)
- Cooper Camp, Dartmouth'26, lab assistant (1 term)
- Keller Bailey, Dartmouth'27, lab assistant (1 term)
- Blaine Boxwell, Dartmouth'27, lab assistant (2 terms)
- Aleaokalani Kahele, Dartmouth'27, lab assistant (2 terms)
- Margaret Raffo, Dartmouth'28, research and lab assistant, studying radionuclides (2 terms)

CONFERENCE ABSTRACTS

- 2026 Hudak, A., et al. (including **Chalif, J.**), 2026. **Fidelity and stratigraphy of the Antarctic Allan Hills old ice archive from Continuous Flow Analysis**. EGU General Assembly 2026 Session CL1.2.7, Vienna, Austria.
- 2025 Brooks, H., Handley, M., Kreutz, K., **Chalif, J.**, Osterberg, E., Jongebloed, U., Winski, D., Kirkpatrick, L., Koffman, B., Wake, C., Schwikowski, M., Skelton, E., Ferris, D., 2025. **Increasing Complexity of Asian Pb Emissions Throughout Industrialization (800 to 2022 CE) as Revealed in the Denali Ice Core Records, Begguya, Alaska**. AGU 2025 Fall Meeting, New Orleans, LA. (Poster)
- Chalif, J.**, Osterberg, E., Winski, D., Kehrwald, N., Lonergan, M., Guiterman, C., Kaplan, J., Jongebloed, U., Youth, E., Edwards, R., Leung, M., Schachterle, M., Jasmann, J., Dibb, J., Scheuer, E., Macken, A., Hantson, S., Kirkpatrick, L., Ferris, D., Handley, M., Kreutz, K., Saltzman, E., Wake, C., 2025. **Atmospheric chemistry implications from Denali ice core fire measurements**. ICECAP Workshop, Seattle, WA. (Talk)
- Jongebloed, U., **Chalif, J.**, Tashmim, L., Porter, W., Bates, K., Chen, O., Osterberg, E., Koffman, B., Winski, D., Kreutz, K., Ferris, D., Wake, C., Alexander, B., 2024. **Dimethyl sulfide oxidation over the industrial era: comparison of long-term observations and several chemical mechanisms**. ICECAP Workshop, Seattle, WA. (Talk)
- Koffman, B., **Chalif, J.**, et al., 2025. **Local versus long-distance dust provenance in a 2019 shallow core at Mount Hunter, Alaska**. 4th Annual US Ice Core Community Meeting, Minneapolis, MN. (Poster)
- Kirkpatrick, L., **Chalif, J.I.**, et al., 2025. **Updates on coordinated ECM and multiproxy sampling in Allan Hills ice**. 4th Annual US Ice Core Community Meeting, Minneapolis, MN. (Talk)
- Chalif, J.**, Osterberg, E., Winski, D., Kehrwald, N., Lonergan, M., Guiterman, C., Kaplan, J., Jongebloed, U., Youth, E., Edwards, R., Leung, M., Schachterle, M., Jasmann, J., Dibb, J., Scheuer, E., Macken, A., Hantson, S., Kirkpatrick, L., Ferris, D., Handley, M., Kreutz, K., Saltzman, E., Wake, C., 2025. **A Siberian fire proxy from the Denali ice core**. 4th Annual US Ice Core Community Meeting, Minneapolis, MN. (Talk)
- 2024 Koffman, B., Jongebloed, U., **Chalif, J.**, Andreasen, J., Fudge, T.J., 2024. **ICECREW: Connecting and Training Icy Early Career Researchers with an Annual Workshop**. AGU 2024 Fall Meeting, Washington, DC. (Poster)
- Choi, A., Peterson, J., Kalk, M., Kirkpatrick, L., Fudge, T.J., **Chalif, J.**, Tipton, L., Davidge, L., Sliwinski, M., Osterberg, E., Koffman, B., Steig, E., Higgins, J., Shackleton, S., Brook, E., 2024. **A High Resolution Look at Carbon Dioxide and Methane Variations in ALHIC1901**. AGU 2024 Fall Meeting, Washington, DC. (Poster)
- Jongebloed, U., **Chalif, J.**, Tashmim, L., Porter, W., Bates, K., Chen, O., Osterberg, E., Koffman, B., Winski, D., Kreutz, K., Ferris, D., Wake, C., Alexander, B., 2024. **Dimethyl sulfide oxidation over the industrial era: comparison of long-term observations and several chemical mechanisms**. AGU 2024 Fall Meeting, Washington, DC. (Poster)
- Brooks, H., Handley, M., Kreutz, K., **Chalif, J.**, Osterberg, E., Jongebloed, U., Winski, D., Kirkpatrick, L., Koffman, B., Wake, C., Schwikowski, M., Skelton, E., Ferris, D., 2024. **Novel lead (Pb) record tracing the impact of human policy and culture on Asian pollution emissions in the North Pacific atmosphere over the last 1,600 years**. AGU 2024 Fall Meeting, Washington, DC. (Poster)
- Osterberg, E.C., **Chalif, J.I.**, Winski, D., Kehrwald, N., Lonergan, M., Youth, E., Kreutz, K., Wake, C., Ferris, D., Edwards, R., Dibb, J., Saltzman, E., Hantson, S., 2024. **North Pacific Wildfire Activity over the past Millennium from a Suite of Alpine Ice Cores**. AGU 2024 Fall Meeting, Washington, DC. (Poster)
- Jongebloed, U.A., **Chalif, J.I.**, Tashmim, L., Porter, W.C., Bates, K.H., Chen, Q., Osterberg, E.C., Koffman, B.G., Cole-Dai, J., Winski, D.A., Ferris, D.G., Kreutz, K.J., Wake, C.P., Alexander, B. **Comparison of four DMS oxidation mechanisms with long-term observations of DMS oxidation products indicate missing DMS chemistry**. GEOS-Chem Meeting, Chicago, IL. (Poster)
- Chalif, J.I.**, Osterberg, E.C., Kreutz, K., Winski, D., Wake, C., Ferris, D., Koffman, B., Kirkpatrick, L., Handley, M., Introne, D., 2024. **Integrity of North Pacific Ice Core Climate Archives in a Warming World**. 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Poster)
- Brooks, H., Handley, M., Kreutz, K., Winski, D., **Chalif, J.**, Osterberg, E.C., 2024. **Echoes from the Arctic: Unravelling Asia's Past Through Pb Fluctuations in Alaskan Ice Cores**. 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Talk)
- Jongebloed, U.A., **Chalif, J.I.**, Tashmim, L., Porter, W.C., Bates, K.H., Chen, Q., Osterberg, E.C., Koffman, B.G., Cole-Dai, J., Winski, D.A., Ferris, D.G., Kreutz, K.J., Wake, C.P., Alexander, B. **Comparison of four DMS oxidation mechanisms with long-term observations of DMS oxidation products indicate missing DMS chemistry**. 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Talk)
- Kirkpatrick, L., **Chalif, J.I.**, et al., 2024. **Coordinated ECM, Water Isotope, Chemistry, and Dust Measurements Provide New Insights into Allan Hills Layering**. 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Talk)
- Winski, D., **Chalif, J.I.**, et al., 2024. **A North Pacific ice core record of summer climate and wildfire history during the last 1000 years**. 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Talk)

Tipton, L., Osterberg, E.C., Koffman, B., Benello, M., **Chalif, J.**, Anderson, K., Chesler, A., Ferris, D., Kreutz, K., Cole-Dai, J., 2024. **Causes of the Late Holocene Dust Spike in Antarctic Ice Cores.** 3rd Annual US Ice Core Open Science Meeting, Portland, ME. (Poster)

Chalif, J.I., Jongebloed, U., Osterberg, E.C., Koffman, B., Alexander, B., Winski, D., Polashenski, D., Stamieszkin, K., Ferris, D., Kreutz, K., Wake, C., Cole-Dai, J., 2024. **Pollution drives enigmatic decline in subarctic biogenic sulfur.** 52nd International Arctic Workshop, Amherst, MA. (Poster)

Brooks, H., Handley, M., Kreutz, K., Winski, D., **Chalif, J.**, Osterberg, E.C., 2024. **Novel lead (Pb) isotope record tracking the impact of policy and the Covid-19 pandemic on Asian pollution emissions deposited in Alaskan ice cores.** 52nd International Arctic Workshop, Amherst, MA. (Talk)

2023 Kirkpatrick, L., Fudge, T.J., **Chalif, J.**, Tipton, L., 2023. **Stratigraphy in highly thinned ice cores.** AGU 2023 Fall Meeting. (Talk)

Jongbloed, U., **Chalif, J.**, Tashmim, L., Porter, W.C., Alexander, B., 2023. **Comparison of ice core vs. modeled changes in DMS oxidation chemistry over the Industrial Era.** AGU 2023 Fall Meeting. (Poster)

Macken, A., Osterberg, E.C., Winski, D., **Chalif, J.**, and others, 2023. **Climatological and Topographical Drivers of Wildfire Growth in Alaska: A Remote Sensing Analysis of 14 Wildfire Events (2001-2019).** AGU 2023 Fall Meeting. (Poster)

Brooks, H., Handley, M., Kreutz, K., Winski, D., **Chalif, J.**, Osterberg, E.C., Kindstedt, I., Erwin, E., Braddock, S., Kirkpatrick, L.R., Wake, C., Kurbatov, A., Gerbi, C., 2023. **Evaluation of public policy and Covid-19 pandemic impacts on historic Asian pollution emissions using Alaskan ice core lead data.** AGU 2023 Fall Meeting. (Poster)

Tipton, L., Osterberg, E.C., Koffman, B., Kurbatov, A., **Chalif, J.**, Adelman, K., Anderson, K., Chesler, A., Ferris, D., Kreutz, K., Cole-Dai, J., 2023. **Causes of the Late Holocene Dust Spike in Antarctic Ice Cores.** AGU 2023 Fall Meeting. (Poster)

Chalif, J., Jongebloed, U., Osterberg, E.C., Koffman, B., Alexander, B., Polashenski, D., Winski, D., Kreutz, K., Wake, C., Stamieszkin, K., Ferris, D., 2023. **Anthropogenic NO_x emissions, not declining marine primary production, drive reduced modern MSA levels in the Denali ice core.** 2nd Annual US Ice Core Open Science Meeting, Seattle, WA. (Talk)

Jongebloed, U., **Chalif, J.**, Larrick, C., Cole-Dai, J., Salimi, S., Edouard, S., Geng, L., Osterberg, E.C., Alexander, B., 2023. **Industrial-Era decline in Arctic methanesulfonic acid is offset by increased biogenic sulfate aerosol.** 2nd US Ice Core Open Science Meeting, Seattle, WA. (Talk)

Tipton, L., Osterberg, E.C., Koffman, B., Kurbatov, A., **Chalif, J.**, Adelman, K., Anderson, K., Chesler, A., Ferris, D., Kreutz, K., Cole-Dai, J., 2023. **What caused the Late Holocene Dust Increase in Antarctic Ice Cores?** 2nd US Ice Core Open Science Meeting, Seattle, WA. (Poster)

2022 Osterberg, E.C., **Chalif, J.**, Koffman, B., Polashenski, D., Winski, D., Kreutz, K., Wake, C., Stamieszkin, K., Ferris, D., 2022. **Modern Decline in North Pacific Ocean Primary Production is Unprecedented for the Past Millennium in the Denali Ice Core.** AGU 2022 Fall Meeting. (Talk)

Youth, E., Osterberg, E.C., Lonergan, M., **Chalif, J.**, Winski, D., Kreutz, K., Wake, C., Ferris, D., Saltzman, E., 2022. **1000 Years of Wildfire Activity in the North Pacific from Three Ice Cores.** AGU 2022 Fall Meeting. (Poster)

Brooks, H., Handley, M., Kreutz, K., Winski, D., **Chalif, J.**, Osterberg, E.C., Kindstedt, I., Erwin, E., Braddock, S., Kirkpatrick, L.R., Wake, C., Kurbatov, A., Gerbi, C., 2022. **Evaluating Asian Policy- and Pandemic-Related Emissions Impacts Using Ice Core Pb Concentration and Isotope Ratio Data from Mt. Hunter (Begguya) 800 – 2022 CE.** AGU 2022 Fall Meeting. (Poster)

Chalif, J., Osterberg, E.C., Winski, D., Kreutz, K., Edwards, R., Wake, C., Dibb, J., Scheuer, E., Saltzman, E., Hantson, S., Kehrwald, N., Lonergan, M., 2022. **A North Pacific Paleofire Proxy from the Denali Ice Core.** 3rd Open Science Meeting of the International Partnerships in Ice Core Sciences, Crans-Montana, Switzerland. (Poster)

Lonergan, M., Osterberg, E.C., Winski, D., Kreutz, K., Wake, C., **Chalif, J.**, Ferris, D., 2022. **Wildfire Activity in the Denali Ice Core Catchment.** 1st US Ice Core Open Science Meeting, La Jolla, CA. (Talk)

Chalif, J., Osterberg, E.C., Winski, D., Kreutz, K., Edwards, R., Wake, C., Dibb, J., Scheuer, E., Saltzman, E., Hantson, S., Kehrwald, N., Lonergan, M., 2022. **Updates on a North Pacific Paleofire Proxy from the Denali Ice Core.** 1st US Ice Core Open Science Meeting, La Jolla, CA. (Poster)

2021 **Chalif, J.**, Osterberg, E.C., Partridge, T., 2021. **Effects and implications of enhanced jet stream waviness over North America during the 1950s-1970s.** AGU 2021 Fall Meeting, virtual. (Poster)